

Project 5 Scope of Work

By: Hunter Anciaux and Elizabeth Wiggins

Date submitted for review: 27 February 2026

Submitted to: Risk Averse, LLC

Goals of Project:

The goal of the project is to analyze the risk of a specific natural hazard, hurricanes, among 6 different states, Texas, Louisiana, Mississippi, Alabama, Georgia, and Florida, and the impact of the hazard. The analysis is to consist of a comparison of how the risk of hurricanes vary across these 6 different states, a report of any trends that occurred in the data, what these trends may indicate about the region, and a discussion of how this natural hazard has impacted the communities of these areas.

Tasks:

Task 1 – Extract and organize SVI data

Data from the Social Vulnerability Index will be downloaded as a CSV. This data will contain information for the entirety of the United States. The values corresponding to Texas, Louisiana, Mississippi, Alabama, Georgia, and Florida will be kept, while the rest will be trimmed, as it does not pertain to the research region.

Task 2 – Preliminary Research

Information pertaining to the project will be gathered in the initial phase. The client requested to use the National Risk Index (NRI) definition for risk, so this must be found beforehand. It would also be useful to understand more about the area and about the impact of hurricanes on the populations of those states.

Task 3 – Examine and Evaluate Risk

Various functions will be used to filter and sort the data related to the hazard in each of the observed states. Areas that rise above the NRI threshold for risk will be examined and

recorded. Other functions, such as the descriptive statistics tool, can be utilized to display values like the mean, median, minimum, and maximum, both in each region and in its entirety.

Task 4 – Plot Data

The hazard data will be plotted to compare it state-by-state, likely on a map-based graph. It can also be put on other charts to show trends and potential correlations between the hazard and its impacts and population, as requested by the client. The regional trends can be visually observed from the charts, namely a map if that is the best format.

Task 5 – Report and Discuss Data

A report will be written containing the information requested by the client. The presence and effects of the risk in each of the 6 states will be compared and displayed in the written report. Here is also where any discovered trends and/or correlations will be discussed.

Project Deliverables:

All files to be submitted by May 8th, 2026, at 5:00pm CT

1. Raw data files used in the Python Code file
2. Python code file used to generate solution
3. Scope of work
4. Gantt Chart
5. Engineering Timesheet
6. Annotated code document (ACD)
7. Written report
8. Link to GitHub Repository